Highlights

Beyond 1-NN Accuracy: Objective-Informed Benchmarking of Time Series Similarity Measures

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- We benchmark time-series measures for prediction, clustering, and separability
- Across 81 datasets, time-series measure rankings change by task
- 1-NN accuracy rankings transfer poorly to downstream tasks
- Matthews correlation coefficient yields clearer measure rankings than accuracy

Beyond 1-NN Accuracy: Objective-Informed Benchmarking of Time Series Similarity Measures

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ABSTRACT

Choosing a time series similarity (or distance) measure can strongly shape the performance and reliability of *learning algorithms*. However, most benchmarking approaches rank measures with a single diagnostic—typically leave-one-out 1-Nearest Neighbor (1-NN)—implicitly generalizing the rankings to other downstream tasks. We ask an external-validity question: do measure rankings obtained under 1-NN persist when the same measures drive other algorithms? To test this, we benchmark a set of measures over three algorithmic families: supervised classification, unsupervised partitioning, and structural separability. We utilize the Friedman-Nemenyi pipeline over 81 binary-label datasets to demonstrate that 1-NN accuracy rankings provide poor generalizations. Additionally, we find that Matthews Correlation Coefficient consistently yields larger between-measure score dispersion across datasets, as measured by statistical power in the Friedman-Iman-Davenport test. Across diagnostics, measure rankings are strongly objective-dependent, with winner groups shifting as the diagnostic changes. Together, these findings caution against treating 1-NN accuracy as a surrogate for "best" distance.

1. Introduction

Time series data is significant for practical and computational problems in fields such as biology, healthcare, economics, industry, and weather planning (Caldeira, Cordeiro, Ruiz and Santos, 2025; Hess and Feuerriegel, 2024; Wald, Goldstein, Efroni, van Amsterdam and Ranganath, 2025; Wu, Qiu, Li, Wang, Hu, Guo, Xiong and Yang, 2024; Rohbeck, De Brouwer, Bunne, Huetter, Biton, Chen, Regev and Lopez, 2025; Briscoe, Gebremedhin, Holder and Cook, 2022; Steinberg, Fries, Xu and Shah, 2023). Similarity (or distance) measures are critical in analyses of algorithms, loss functions, or benchmarking objectives, yet their selection is too often asserted rather than justified. The assertions are implicitly traditional adherence or some rule-of-thumb instead of being based on research in ranking measures (measure benchmarks). While insightful, we find the measure benchmark literature is often based on a single classifier–this is dominantly "leave-one-out" 1-nearest-neighbor classification (1-NN).

Measure benchmarking parallels classifier benchmarking. Demšar (2006) ranks classifiers of tabular datasets, and Bagnall, Lines, Bostrom, Large and Keogh (2017) extends Demšar's work to ranking time series classifiers. Both works are driven by accuracy and use the Friedman-Nemenyi pipeline—a pipeline that checks if a set of rankings indicates statistical significance over order (the Friedman test), then conducts pairwise comparisons to derive statistically indistinguishable groups (Nemenyi post-hoc analysis). Later, Liu and Paparrizos (2024) ranks time-series algorithms by their anomaly-detection abilities.

In 2025, Shao et al. seek to unify time-series forecasting benchmarking under a standard pipeline, suggesting that dataset heterogeneity causes conflicting results in the literature. Similarly, Ragab, Eldele, Tan, Foo, Chen, Wu, Kwoh and Li (2023) seek to standardize evaluation through ADATIME, a benchmarking suite for time-series domain adaptation.

Paparrizos, Liu, Elmore and Franklin (2023) offer a recent time series measure benchmark. They conduct accuracy and runtime analyses of time series distance measures partitioned by measure type (e.g., lock-step, elastic, kernel). Here, they use 1-NN to find KDTW (a kernel variant of dynamic time warping) to have the highest accuracy and cost.

Wang, Mueen, Ding, Trajcevski, Scheuermann and Keogh (2013) perform a similar benchmark comparing accuracies with the 1-NN classifier, but without the Friedman-Nemenyi pipeline. They discover that elastic distances best lock-step distances on small datasets and that a window-constrained DTW is a safe default. Wang et al. found no

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universally superior measure–but found it to be dataset dependent. In their future work, Paparrizos et al. (2023) echo the belief that the best time series measure is data dependent. Our results reinforce this conclusion.

Recent work in measures for multivariate time series settings often preserves the same evaluation proxy of 1-NN with accuracy. For example, Shifaz, Pelletier, Petitjean and Webb (2023) contribute multivariate variants of several elastic measures. This work suggests that the measures support many applications such as clustering, classification, anomaly detection, indexing, subsequence search, and segmentation, yet the empirical comparisons are still anchored to classification–especially the one-nearest neighbor classifier.

Dau, Bagnall, Kamgar, Yeh, Zhu, Gharghabi, Ratanamahatana and Keogh (2018) visualize cross-dataset results with *critical difference diagrams* (CDDs) derived from 1-NN classifier accuracy. CDDs visualize average rankings with a black bar that groups rankings as statistically equivalent. A central caution in this work is the Texas-sharpshooter fallacy: retroactively framing a method as "for this subset of datasets" after observing outcomes, thereby overstating generality or drawing post-hoc domain boundaries. However, using just one diagnostic also risks overstating generality, as we show in our analyses. We design our framework to guard against such statements.

Across this literature, the common way to ground *"which measure is best?"* is treating the measure quality as 1-NN classification performance, evaluated by accuracy and using the leave-one-out method. Later works (Paparrizos et al., 2023; Górecki, Łuczak and Piasecki, 2024) combine the 1-NN method with statistical rigors such as the Friedman test and Nemenyi's post-hoc analysis. The Friedman test is *global* and asks if the ranking differences are significant, while Nemenyi's post-hoc analysis is *pairwise*—asking if this measure's rank significantly differs from another's. The appeal for 1-NN is pragmatic: it is parameter free, contains no training biases, and leverages the entire dataset (besides the test data point). However, the intuition for using the 1-NN diagnostic deserves scrutiny: does label-matching the closest datum really indicate a "good" measure? This diagnostic can convey if a measure has label-informative value; however, only considering the closest neighbor biases to micro-scale geometry. Perhaps a more robust assessment requires evaluating measures across multiple algorithmic families that probe different scales of structure—including local neighborhoods, global partitions, and density-based separations. Our work is centered around the question: "does 1-NN with accuracy yield similar results as other diagnostics, such as spectral clustering or a support-vector classifier?" It makes the following contributions:

- We propose objective-informed benchmarking across 3 diagnostic families.
- We find that 1-NN with accuracy has poor generalizability to other diagnostics.
- We show that Matthews Correlation Coefficient (MCC) consistently yields higher rank-discrimination power than accuracy, using empirical tests on 81 datasets from a wide spectrum of applications.
- We present the top-performing measures and the critical difference diagrams for each diagnostic. We find that distance-measure rankings are strongly objective-dependent.

2. Background

We discuss the selection of time series measures and datasets included in our empirical analysis. Here, we also introduce the relevant rank matrix that is passed into the Friedman-Nemenyi pipeline.

2.1. Measures

We evaluate seventeen measures selected for their diversity in geometric assumptions. These measures are organized into five families:

- L_p *Minkowski*: captures standard norm-based geometries.
- L_1 *distance*: various normalizations for absolute differences.
- *Squared-chord*: assumes data resembles an empirical probability distribution.
- *Squared* L_2: Euclidean variants with scaling, normalizing, or censoring adjustments.
- *"Other"*: four miscellaneous measures.

Family	Abbrev.	Metric	Function $M(a,b)$ for $a,b \in \mathbb{R}^d$
Lp Minkowski	MD	Manhattan distance	$\sum_{i=1}^{d} \lvert a_i - b_i \rvert$
	CD	Chebyshev distance	$\max_{1 \le i \le d} \lvert a_i - b_i \rvert$
	ED	Euclidean distance	$\sqrt{\sum_{i=1}^{d} (a_i - b_i)^2}$
L_1 distance	LD	Lorentzian distance	$\sum_{i=1}^{d} \ln\left(1 + \lvert a_i - b_i \rvert\right)$
	CanD	Canberra distance	$\sum_{i=1}^{d} \frac{\lvert a_i - b_i \rvert}{\lvert a_i \rvert + \lvert b_i \rvert}$
	SoD	Soergel distance	$\frac{\sum_{i=1}^{d} \lvert a_i - b_i \rvert}{\sum_{i=1}^{d} \max(a_i, b_i)}$
	MCD	Mean Character distance	$\frac{1}{d} \sum_{i=1}^{d} \lvert a_i - b_i \rvert$
	NID	Non Intersection distance	$\frac{1}{2} \sum_{i=1}^{d} \lvert a_i - b_i \rvert$
Squared-chord	MatD	Matusita distance	$\sqrt{\sum_{i=1}^{d} \left(\sqrt{a_i} - \sqrt{b_i}\right)^2}$
	HeD	Hellinger distance	$\sqrt{2 \sum_{i=1}^{d} \left(\sqrt{a_i} - \sqrt{b_i}\right)^2}$
Squared L_2	ClaD	Clark distance	$\sqrt{\sum_{i=1}^{d} \left(\frac{a_i - b_i}{\lvert a_i \rvert + \lvert b_i \rvert}\right)^2}$
	AD	Average (Euclidean) distance	$\sqrt{\frac{1}{d} \sum_{i=1}^{d} (a_i - b_i)^2}$
	MCED	Mean Censored Euclidean distance	$\sqrt{\frac{\sum_{i=1}^{d} (a_i - b_i)^2}{\sum_{i=1}^{d} \mathbf{1}\left[a_i^2 + b_i^2 \neq 0\right]}}$

Table 1
Measures used in this work, grouped by family.

Abbrev.	Metric	Function $M(a,b)$ for $a,b \in \mathbb{R}^d$
JacD	Jaccard distance	$\frac{\sum_{i=1}^{d} (a_i - b_i)^2}{\sum_{i=1}^{d} a_i^2 + \sum_{i=1}^{d} b_i^2 - \sum_{i=1}^{d} a_i b_i}$
HamD	Hamming distance	$\sum_{i=1}^{d} \mathbf{1}\left[a_i \neq b_i\right]$
WIAD	Whittaker's index of association	$\frac{1}{2} \sum_{i=1}^{d} \left\lvert \frac{a_i}{\sum_{k=1}^{d} a_k} - \frac{b_i}{\sum_{k=1}^{d} b_k} \right\rvert$
HasD	Hassanat distance	$\sum_{i=1}^{d} \begin{cases} 1 - \frac{1 + \min(x,y)}{1 + \max(x,y)}, & \min(x,y) \geq 0, \\ 1 - \frac{1 + \min(x,y) + \lvert \min(x,y) \rvert}{1 + \max(x,y) + \lvert \min(x,y) \rvert}, & \min(x,y) < 0. \end{cases}$

Table 2
Other measures used in this work.

We provide full details (names, abbreviations, and formal definitions) in Tables 1 and 2.

Since our experiments are computationally expensive, we restrict our chosen measures to those with a runtime exactly Θ(number of time steps). Our goal is to find the generalizability of the 1-NN classifier based on an accuracy diagnostic, so this selection is acceptable and currently necessary. Including measures such as dynamic time warping is an interesting direction for future work. For measures that are undefined for identical segments, we insert a distance of zero when identical segments are seen.

2.2. Datasets

We evaluate distance measures on 81 univariate time series datasets. This collection spans diverse application domains including healthcare monitoring (ECG, EMG, EOG), human activity recognition, spectroscopy, environmental sensing, and synthetic benchmarks. The datasets are described in Tables 4 and 5, and their sizes are summarized in Table 3.

Domain	Modality	Datasets	Primary Task
Acoustic	Speech	Phoneme	Phoneme classification from short audio segments
Biomedical	Cardiac	ECG 5000, ECG Five Days, CinC ECG Torso, Non-Invasive Fetal ECG (2)	Cardiac rhythm classification and arrhythmia detection
	EMG/EOG	Semg Hand (3), EOG Signal (2)	Muscle- and eye-movement–based gesture and gaze classification
	Clinical	Diabetes Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, Vanderplas, Passos, Cournapeau, Brucher, Perrot and Duchesnay (2011)	Binary disease-state classification from glucose measurements
Energy	Appliances	Computers, Large Kitchen Appliances, Refrigeration Devices, Screen Type, Power Cons, House Twenty, Freezer (2)	Device-state and appliance identification from power consumption
	Building	Energy Consumption Hebrail and Berard (2012)	Consumption-pattern classification in building energy usage
Environmental	Geophysical	Earthquakes, Lightning (2), Rock, Sunspot Seabold and Perktold (2010)	Event detection and regime classification in geophysical time series
	Biological	Insect Wingbeat Sound, Insect EPG (2), Fungi, Pig Vitals (3)	Species identification and physiological state classification
Infrastructure	Transport	Ford (2), Car, Dodger Loop (3)	Vehicle identification and traffic-pattern classification
	Sensors & HVAC	Mote Strain, Room Temperature Candanedo (2016), Azure Predictive Maintenance Microsoft Azure (2016)	Structural state classification and equipment health monitoring

Table 4
Dataset collection organized by domain and modality (81 univariate time series). Most datasets are obtained from Dau, Bagnall, Kamgar, Yeh, Zhu, Gharghabi, Ratanamahatana and Keogh (2019). For datasets obtained from other sources, the references are provided next to the dataset. Continued in Table 5.

Dataset selection prioritized a sufficient sample size (1K–30K observations) to promote representativeness and reliability for our analyses. All datasets are cast as binary classification problems to ensure methodological consistency across our algorithmic families and their corresponding diagnostic objectives. Regression targets are binarized at the median, while multiclass labels are cast as the majority class or any other class.

Sample Count	Datasets
1K – 2K	16
2K – 5K	30
5K – 10K	16
10K – 20K	13
20K – 30K	6
Total	**81**

Table 3: Dataset characteristics

Preprocessing emphasizes temporal continuity over sample retention. We apply standard missing value imputation with linear interpolation for short gaps, but critically, longer gaps define segment boundaries rather than being imputed—arbitrary interpolation across extended periods destroys the sequential structure required for meaningful similarity computation. After splitting at null boundaries, we retain only sufficiently long segments and apply per-segment min-max normalization (between zero and one; chosen to preclude negative square root errors). This segment-wise approach ensures scale-invariance across diverse measurement units while preserving within-segment temporal dynamics. For distance computation, preprocessed segments are windowed into fixed-length observations of 32 time steps.

2.3. Rank Matrix

Let $\mathcal{M} = \{m_1, \dots, m_j, \dots m_C\}$ be the list of distance measures and $\mathcal{D} = \{D_1, \dots, D_i, \dots, D_B\}$ be the list of datasets. Then for a given algorithm A, we construct a rank matrix $\mathbf{R} \in \mathbb{R}^{B \times C}$ where the ranking of measure m_j given dataset D_i is $1 \leq \mathbf{R}_{i,j} \leq C$.

Domain	Modality	Datasets	Primary Task
Motion & Gesture	Wearable	All Gesture Wiimote (3), Gesture Pebble (2), Pickup Gesture Wiimote Z, Gesture Mid-Air (3)	Gesture classification from wearable and depth-based motion signals
	Action	Gun Point (4), Cricket (3), Inline Skate, Haptics	Human action classification from pose and trajectory sequences
Shape-Based	Trajectory, Visual	Arrow Head, Fish, Plane, Fifty Words, Adiac	Shape and trajectory classification from temporal contours
Socioeconomic	Public, Economic	UK Driver Deaths Seabold and Perktold (2010), US Acc Deaths Seabold and Perktold (2010), California Housing Pedregosa et al. (2011)	Temporal trend classification in public and economic indicators
Spectral	Spectroscopy	Beef, Coffee, Ham, Meat, Olive Oil, Chlorine Concentration, Ethanol Level	Material identification and concentration discrimination from spectra
Synthetic	Benchmarks	ACSF1, BME, CBF, Mallat, Mixed Shapes (2)	Controlled pattern classification under known generative structure

Table 5
Dataset collection organized by domain and modality (81 univariate time series). Most datasets are obtained from Dau et al. (2019). For datasets obtained from other sources, the references are provided next to the dataset. Continued from Table 4.

Rankings are populated by evaluating each measure m_j with a fixed algorithm A on dataset D_i with appropriate performance metric P (e.g., accuracy). A lower rank indicates better performance. Following Garcia and Herrera (2008), ties are handled by average rank; for example, scores of (1, 2, 2, 3) produce rankings of (1, 2.5, 2.5, 4).

When an algorithm such as spectral clustering expects a similarity measure, we map the distance measures with the Gaussian kernel as done in Bodó, Minier and Csató (2011).

2.4. Friedman-Nemenyi Pipeline

The Friedman test is a non-parametric test that operates on the average ranks of the measures. Under this null hypothesis, all measures have the same expected rank across datasets. We adopt the Iman-Davenport refinement of the Friedman test since the original test is undesirably conservative Iman and Davenport (1980); Demšar (2006). Following Demšar's recommendation, we reject the null hypothesis for $p_{\text{Friedman}} \leq 0.10$.

If the Friedman test finds the rankings to be statistically significant, we proceed with Nemenyi's post-hoc analysis. Nemenyi's post-hoc analysis does pairwise comparisons (pairwise over measures) to define a "critical difference" scalar. Measures whose respective ranks are within the range of this critical difference are statistically indistinguishable (Nemenyi, 1963).

3. Methods

We present the best distance measures for several algorithms using the rigorous Friedman-Nemenyi pipeline. This pipeline requires a collection of ranked lists (a ranked matrix) before deciding if the measure rankings are statistically significant, then identifying statistically indistinguishable groupings. We first explain the rank matrix construction and our Friedman-Nemenyi adoption, then we reason performance metric and null-hypothesis significance test choices based on the algorithmic family.

Algorithm 1 articulates our adoption of the Friedman-Nemenyi pipeline. We highlight the creation of the rank matrix with the appropriate performance metric P (corresponding to algorithm A's family), demonstrate the gatekeeping affect of the Friedman test–failure indicates all measures are statistically indistinguishable, then we quantify the critical difference we use in Nemenyi's post-hoc analysis.

All algorithms (diagnostics) we test learn a concept that computes a pairwise distance matrix (structure recording distances between every point pair) and assigns each data point a class, cluster, or scalar value. The nuance of assignment affects the diagnostic's objective, as we summarize in Table 6. We partition these diagnostics into

Algorithm 1 Objective-Informed Friedman-Nemenyi Pipeline

Input: Algorithm A, performance metric P, datasets $\mathcal{D} = \{D_1, \ldots, D_i, \ldots, D_B\}$, measures $\mathcal{M} = \{m_1, \ldots, m_j, \ldots, m_C\}$.
Output: Rank-ordered list of measures that are statistically indistinguishable with the best $\mathcal{M}^*$.
Populate Rank Matrix.
Define the $\mathcal{D} \times \mathcal{M}$ rank matrix as $\mathbf{R} \in [1, C]_{\mathbb{R}}^{B \times C}$.
for each dataset $D_i \in \mathcal{D}$ **do**
    Define the performance cache as $s \in \mathbb{R}^C$.
    for each measure $m_j \in \mathcal{M}$ **do**
        $s_j \leftarrow \mathrm{P}(\mathrm{A}(D_i; m_j))$
    end for
    $\mathbf{R}_i \leftarrow \mathrm{rank}(s)$ ▷ Lower rank is better.
end for
Begin Friedman-Nemenyi Pipeline.
$p_{\text{Fried}} \leftarrow$ IMAN-DAVENPORT-FRIEDMAN-TEST($\mathbf{R}$)
if $p_{\text{Fried}} > 0.10$ **then**
    return $\mathcal{M}$ ▷ All measures are statistically indistinguishable.
end if
$\bar{\mathbf{r}} \in [1, C]_{\mathbb{R}}^{C}$ is the average-rank of m_j over $\mathcal{D}$.
$\mathrm{CD} \leftarrow q_{\alpha,C} \sqrt{\dfrac{C(C+1)}{6B}}$ ▷ Nemenyi's CD uses the Studentized range statistic $q_{\alpha,C}$.
$\mathcal{M}^* \leftarrow \{\, m \in \mathcal{M} : \bar{\mathbf{r}}_j \leq \min \bar{\mathbf{r}} + \mathrm{CD} \}$ ▷ Measures statistically tied with the best.
return $\mathrm{sort}(\mathcal{M}^* \text{ by } \bar{\mathbf{r}}_j \uparrow)$ ▷ Winner group, ordered by average rank.

Family	Diagnostic Objective	Performance Metric	Null Hypothesis Test
Supervised	Prediction–Label Agreement	MCC	McNemar Test
Unsupervised	Partition–Label Agreement	ARI	Fisher Test
Structural	Label–Distribution Separability	AUROC	Mann–Whitney U Test

Table 6
Components of the three diagnostic families used in the Friedman-Nemenyi ranking pipeline.

three families: supervised (1-NN, support-vector classifier), unsupervised (spectral clustering, k-medoids, aggregate clustering), and label-separation (RNLDD, weighted $(n-1)$-NN).

3.1. Supervised Pipeline

The supervised pipeline applies to all interested in supervised classification and estimates a common applicable concern of practitioners: *can this measure provide reliable predictions?* Statistically, this question is better worded as *how strong is the label-prediction agreement?* than *what proportion of predictions are correct?*.

We choose Matthew's Correlation Coefficient (MCC) over accuracy (ACC) since MCC answers the former while ACC answers the latter. The range of MCC values $\in [-1, 1]$, where MCC $= -1$ indicates perfect disagreement, MCC $= 0$ signifies the results are no better than random guessing, and MCC $= 1$ indicates perfect agreement. As supporting evidence of MCC's strengths over accuracy–including disparate majority/minority class sizes, we recommend Chicco, Tötsch and Jurman (2021). For a short intuition, MCC is a rigorously defined coefficient that is uniquely sensitive to perturbations of any confusion-matrix cell, while accuracy is a ratio that is reducible to a binary (correct/incorrect) function (Briscoe, Kepler, Deford and Gebremedhin, 2025).

Since paired disagreement is the null, our complementary null hypothesis test is the McNemar test (McNemar, 1947). For this significance test, our baseline is the majority label.

3.2. Unsupervised Pipeline

The unsupervised pipeline applies whenever the practitioner cares about how well some distance-induced partition of the data aligns with the ground-truth labels, without giving the learning algorithm access to those labels. In other words, instead of asking *"can this measure drive a good classifier?"* (supervised), we ask *"if we cluster in this measure's geometry, do the resulting groups coincide with the labels?"* This may be useful in applications where data comes before the label or when concept drift occurs.

We score partition–label agreement using the chance-corrected clustering metric Adjusted Rand Index (ARI). ARI analyzes pairwise agreements between labels and cluster assignments then rescales to ARI $\in [-1, 1]$ where random partitions have expectation 0, perfect agreement yields ARI $= 1$, and worse-than-random partitions have ARI < 0. Unlike accuracy on clusters, ARI does not require cluster-label assignment and inherently compensates for label imbalance.

For statistical significance, we use the two-sided Fisher's test to check for the null hypothesis—which is that there is no association between the two clusters and two labels being compared (Hubert and Arabie, 1985).

3.3. Structural Pipeline

The structural (label–separation) pipeline applies whenever the practitioner has a scalar score derived from the distance structure (e.g., a density score or a weighted nearest-neighbor score) and cares about how cleanly this score separates the two label classes. Here the question is not *"can we assign a hard class label?"* or *"do clusters match labels?"* but instead *"can the measure induce a distribution on which positive and negative examples occupy distinguishable regions?"*

We quantify label–distribution separability with AUC (Area Under the Curve). Given a scalar score for each instance and a binary label, AUC assesses how well the groups are separated by pair-wise comparisons. It is inherently threshold-free and invariant to any strictly monotone transformation of the scores (such as scaling). In other words, AUC has no specific pivot and cares about order instead of magnitudes. For insight, we review AUC in Equation 1 for classes C_0 and C_1, and structural score function S.

$$\mathrm{AUC}(S) = \frac{\sum_{t_0 \in C_0} \sum_{t_1 \in C_1} \mathbf{1}[S(t_0) < S(t_1)]}{|C_0| \cdot |C_1|} \tag{1}$$

However, our label-separability objective is independent of which class typically yields lower scores—$S(t_0) < S(t_1)$ or $S(t_0) > S(t_1)$ is immaterial. So for label invariance, we symmetrize AUC by the transformation AUC $\leftarrow \max(\mathrm{AUC}, 1 - \mathrm{AUC})$. A value near 0.5 indicates no useful separation, whereas AUC close to 1 indicates strong separation of the two label distributions.

We use the p-value from the Mann-Whitney U test since this null is that the two label distributions are identical and any apparent separation is due to sampling noise (AUC ≈ 0.5). The Mann-Whitney U test is natural since AUC is equivalent to the normalized Mann-Whitney U statistic (Cortes and Mohri, 2003).

3.3.1. Weighted $(n-1)$-NN.

Our large-scale nearest-neighbor diagnostic considers every data point in its neighborhood $\mathcal{N}(k)$ except k which is the point being predicted:

$$\mathcal{N}(k) = \{1, \ldots, n\} \setminus \{k\}.$$

This neighborhood is then partitioned by label $c \in 0, 1$: $\mathcal{N}(k) = \mathcal{N}_0(k) \cup \mathcal{N}_1(k)$. For each class c, we compute the average distance from k to that class,

$$\mathrm{avg}\left(k, \mathcal{N}_c(k)\right) = \frac{1}{|\mathcal{N}_c(k)|} \sum_{g \in \mathcal{N}_c(k)} \mathbf{D}_{k,g}, \tag{2}$$

and define the structural score

$$S_k = \mathrm{avg}\left(k, \mathcal{N}_1(k)\right) - \mathrm{avg}\left(k, \mathcal{N}_0(k)\right). \tag{3}$$

Thus, S_k is the signed margin of k's average distance between the two classes. $S_k < 0$ means k averages closer to class 0.

Weighted (n-1)-NN is useful for scenarios demanding a holistic view of class separability. In our studies, it is useful for insight into the tested measures. For applications, it gives insight for anomaly detection studies such as in cyber security.

3.3.2. RNLDD

Our local density-based diagnostic is Rank-Normalized Local Density Divergence (RNLDD). RNLDD is parameterized by $q \in (0, 1)$, the proportion of data defined as local (the neighborhood). With the pair-wise distance matrix $\mathbf{D}$, nearest-neighbors set $\mathcal{N}(k)$, and population size n, we calculate the local reachability density of point k as ρ_k. Next, we obtain the relative density divergence (S_k) by ranking, where the densest point receives rank one:

$$\rho_k = \frac{|\mathcal{N}_q(k)|}{\sum_{\ell \in \mathcal{N}_q(k)} \mathbf{D}_{k,\ell}}, \quad S_k = \frac{\text{rank}(\rho_1, \ldots, \rho_n)_k - 1}{n - 1}.$$

In our studies, RNLDD gives insight on class separability by local densities. In applications, RNLDD could give information in anomaly detection applications by harnessing the local density manifold to identify deviations from normal patterns. For example, a high-value, fraudulent credit card transaction among a cluster of local, low-value purchases.

3.4. Diagnostic-Dataset Compatibility Check

When the Friedman test fails to reject the null hypothesis ($p_{\text{Friedman}} > 0.10$) for algorithm A, we check if this failure is influenced by algorithm/dataset incompatibility. Such incompatibility occurs when the algorithm cannot draw a boundary better than random guess or than majority-label guessing for any tested measure.

Algorithm A has a p-value for every dataset/measure pair which can check that the given dataset/measure pair is label informative for A. To check if the dataset is label informative for A, we fuse all p-values over the measure dimension using the harmonic mean to control for family-wise error under dependence (Wilson, 2019). Each dataset j then has a corresponding harmonic-mean p-value called HMP_j.

4. Results

We begin by reviewing diagnostic-dataset compatibilities for all algorithms and dataset compatibilities in Figure 1. Intuitively, this graph counts how many datasets were statistically significant in improvement over a trivial baseline such as random guessing or a majority label.

In Figure 2, we see that MCC consistently yields more Friedman statistical power than accuracy. This indicates that MCC provides sharper discrimination between distance measures. Next, we show that 1-NN with MCC yields more statistical power than all other objectives in Figure 3.

Figure 4 illustrates each diagnostic's similarity to the baseline "1-NN with accuracy" by comparing how quick the baseline's top five measures (the "target-bag") are reached over each diagnostic. The x-axis represents the "X" highest ranking measures of the respective diagnostic, and the y-axis is the percent of measures from the target-bag also seen in the diagnostic's "Top-X". For example, "1-NN with MCC" has the coordinate (3,60); this means that 60% of the target-bag's measures are found in this diagnostic's top 3 measures.

We present the rankings by the traditional 1-NN with accuracy in Figure 5. This figure is a "critical difference diagram" (CDD). In a CDD, the average ranking of each measure over all datasets is presented along a ranking dimension, and "critical differences" (obtained via Nemenyi's post-hoc analysis) are shown by black bars.

For the supervised family, we highlight the critical difference diagrams for 1-NN and Support Vector Classifier in Figure 6. Both algorithms are assessed with MCC over "leave-one-out" cross validation. For the unsupervised family, we present the CDD for K-Medoids in Figure 7, the CDD for aggregate clustering in Figure 8, and the CDD for spectral clustering in Figure 9. For the structural family, we present RNLDD and weighted (n-1)-NN in Figure 10.

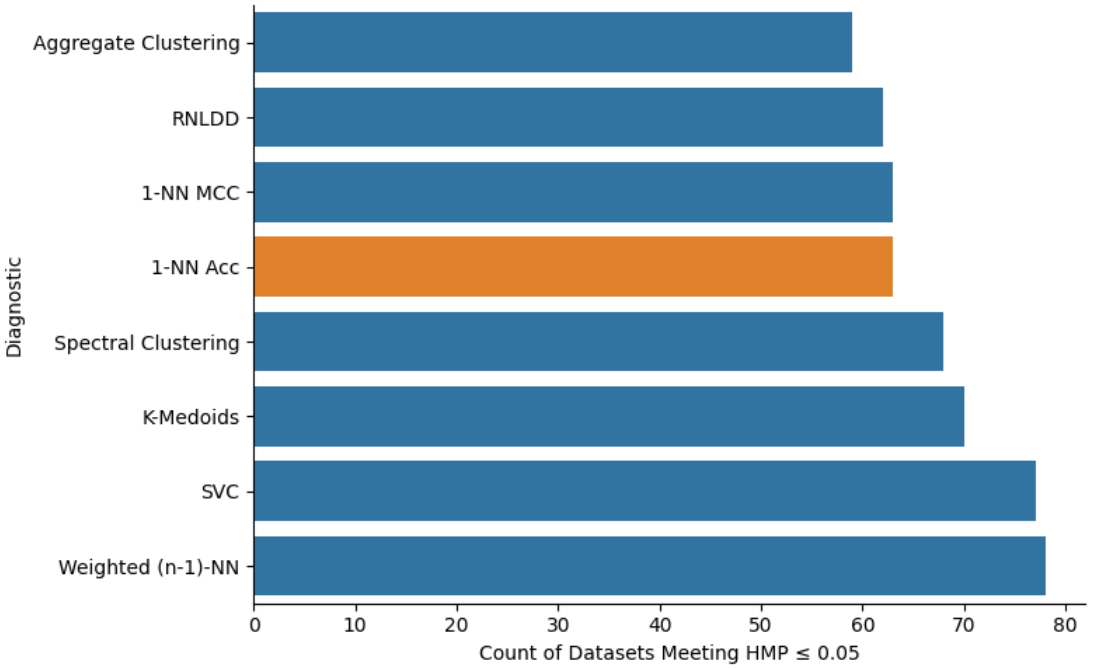


Figure 1: Comparing dataset compatibility for each diagnostic. We assess by comparing each dataset's harmonic mean p-value (HMP) to 0.05.

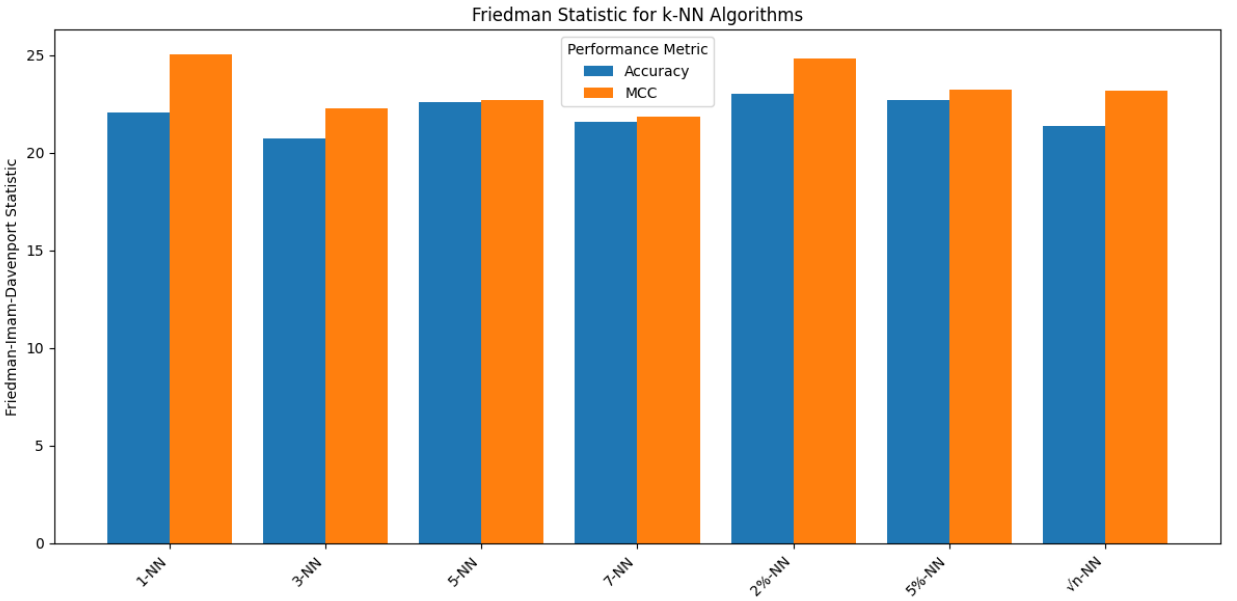


Figure 2: Comparison of Friedman test outcomes for k-NN diagnostics for $k \in \{1, 3, 5, 7, 0.02n, 0.05n, \sqrt{n}\}$. MCC consistently has more statistical power than accuracy.

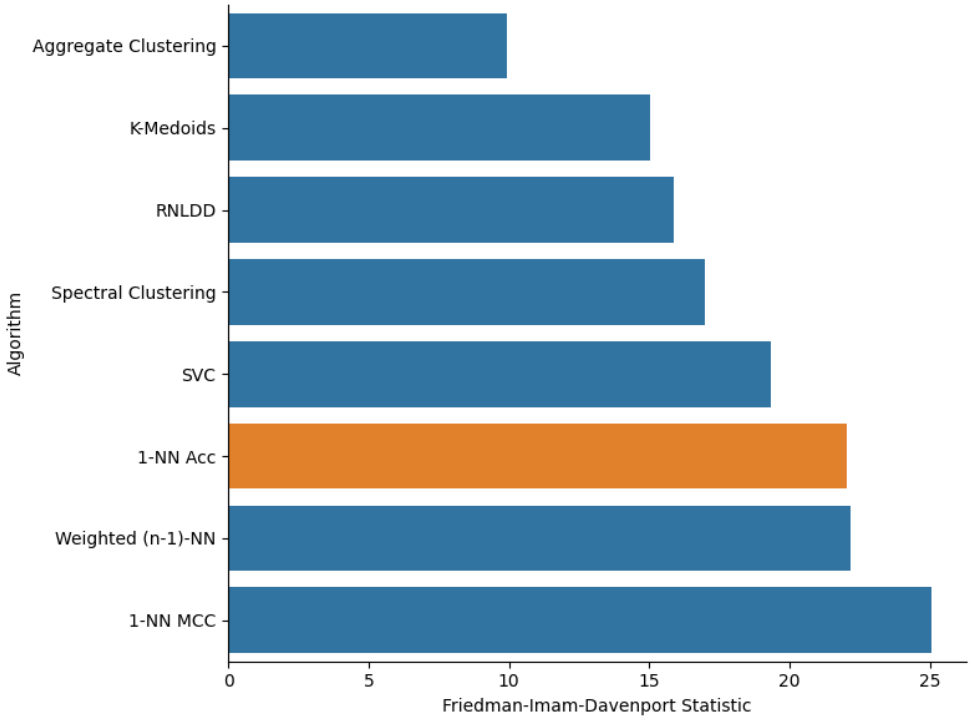


Figure 3: Comparison of Friedman-Imam-Davenport statistics for various diagnostics.

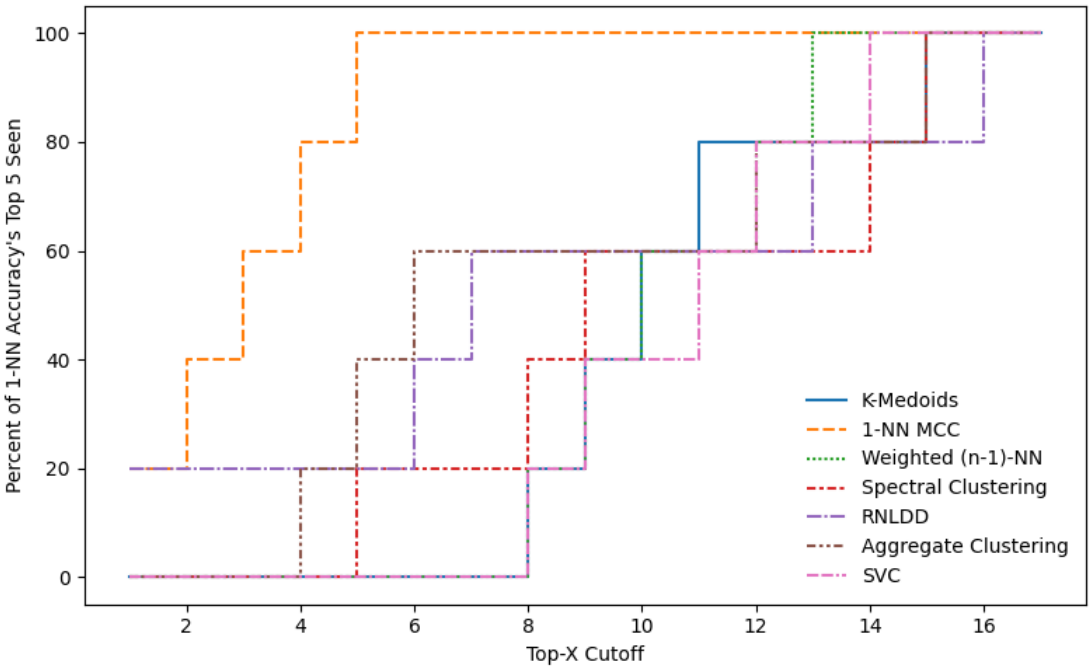


Figure 4: Top-X overlap with the top five measures from the 1-NN with Accuracy diagnostic baseline. The curve shows the fraction of measures in the baseline's top 5 that also appears within the given diagnostic's top-X measures. Lines that rise earlier (high overlap at small X) indicate stronger agreement with the baseline, while late-rising curves indicate poorer transferability from the baseline's top rankings.

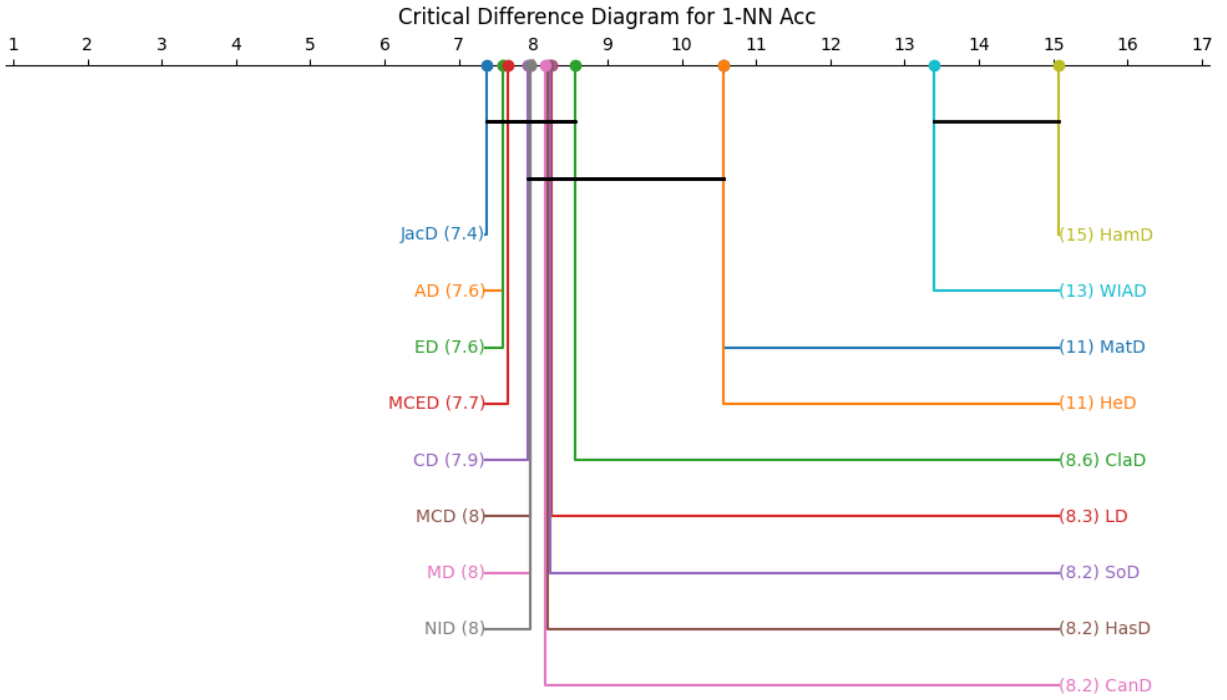
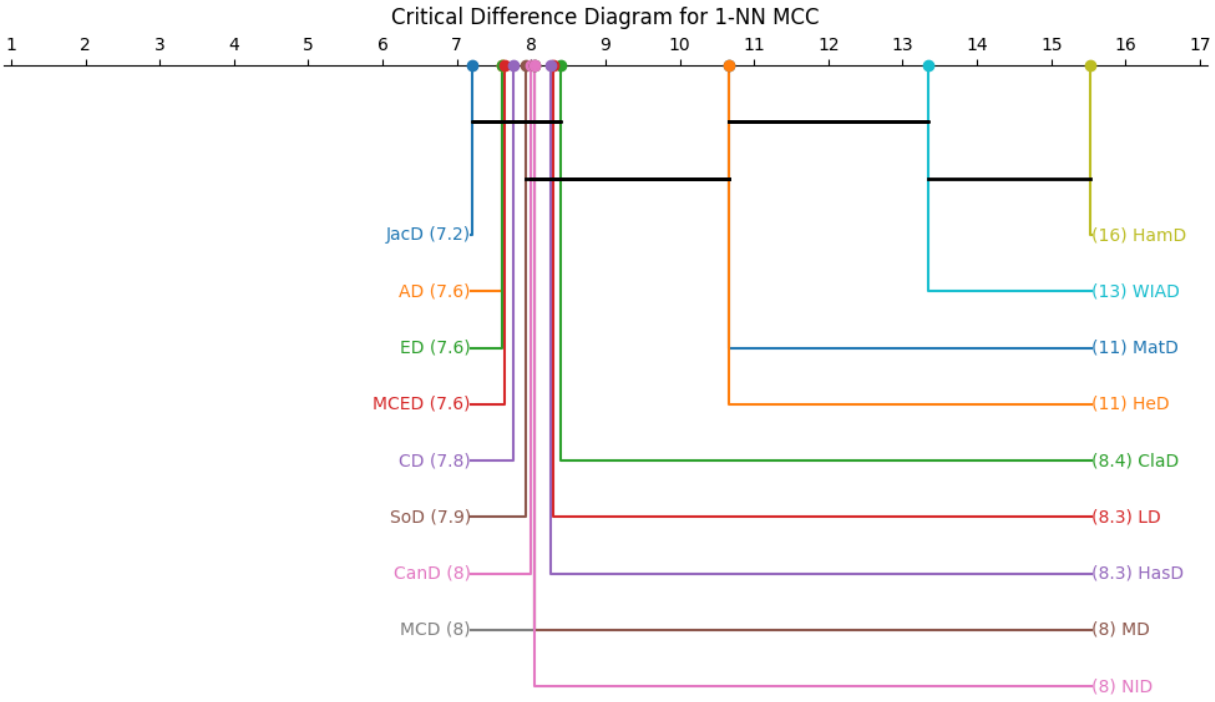
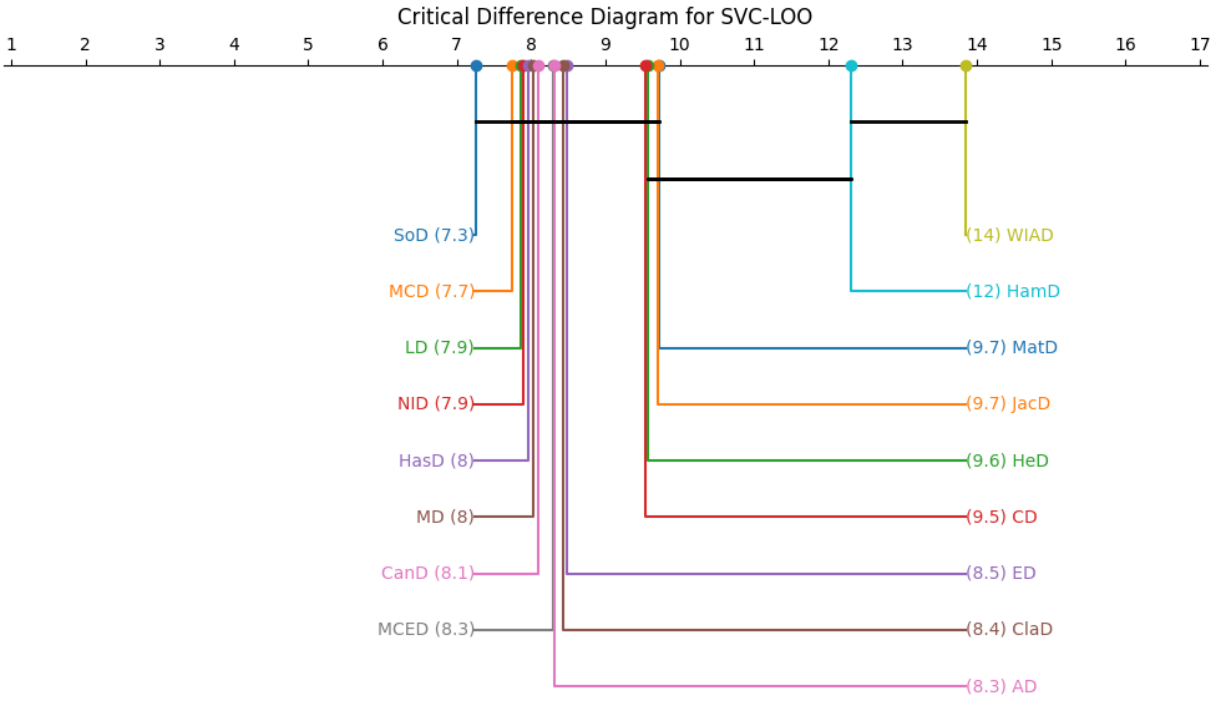


Figure 5: The Critical Difference Diagram for the standard 1-NN with Accuracy Diagnostic. The top 13 measures are statistically indistinguishable per Nemenyi's critical difference.



(a) Leave-One-Out 1-Nearest Neighbor with MCC Diagnostic



(b) Leave-One-Out Support Vector Classifier with MCC Diagnostic

Figure 6: The critical difference diagrams for two supervised diagnostics using the MCC performance metric. Each subfigure shows the average measure rank over 81 datasets. The black bars are statistically indistinguishable groups.

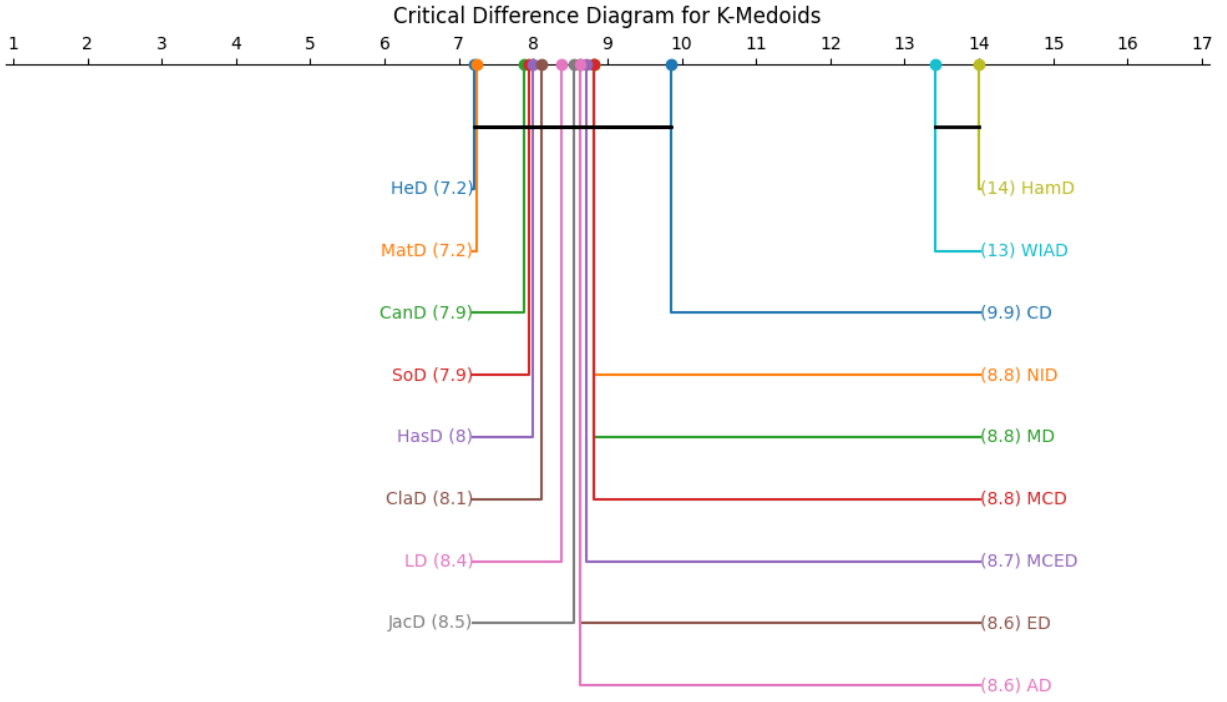


Figure 7: The critical difference diagram for the "k-medoids with ARI" diagnostic. Each measure's average rank over 81 datasets is given. The black bars are statistically indistinguishable groups.

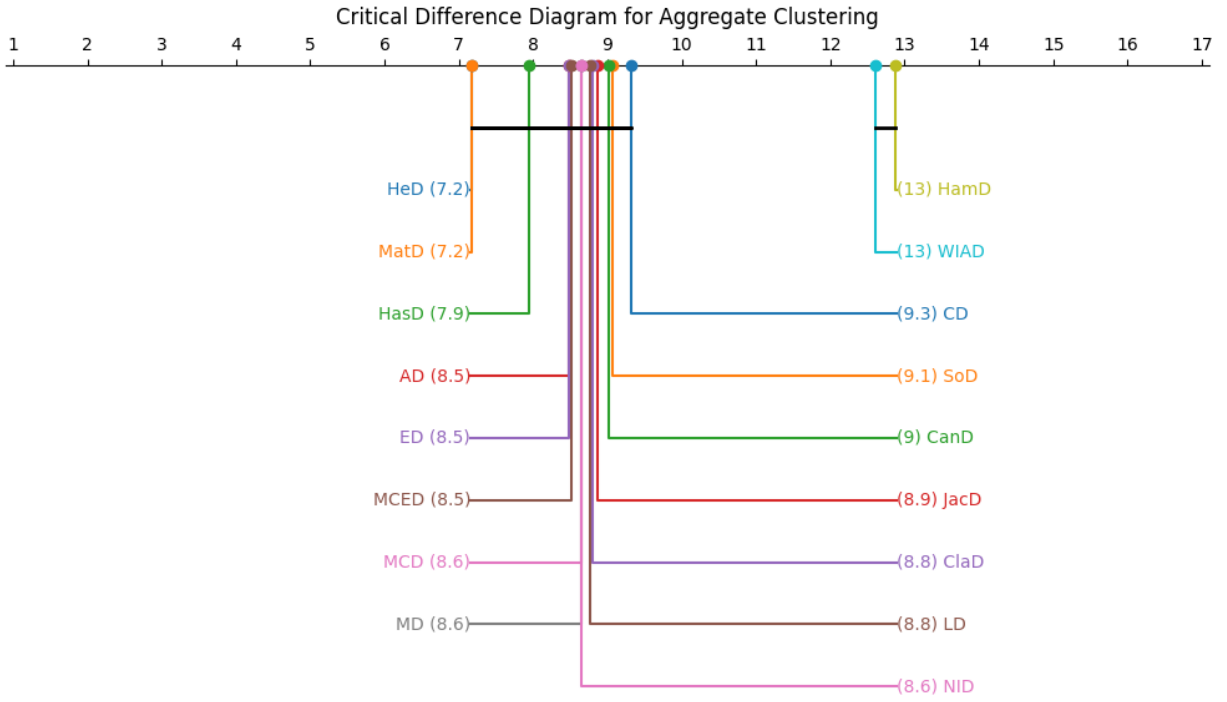


Figure 8: The critical difference diagram for "aggregate clustering with ARI" diagnostic. Each measure's average rank over 81 datasets is given. The black bars are statistically indistinguishable groups.

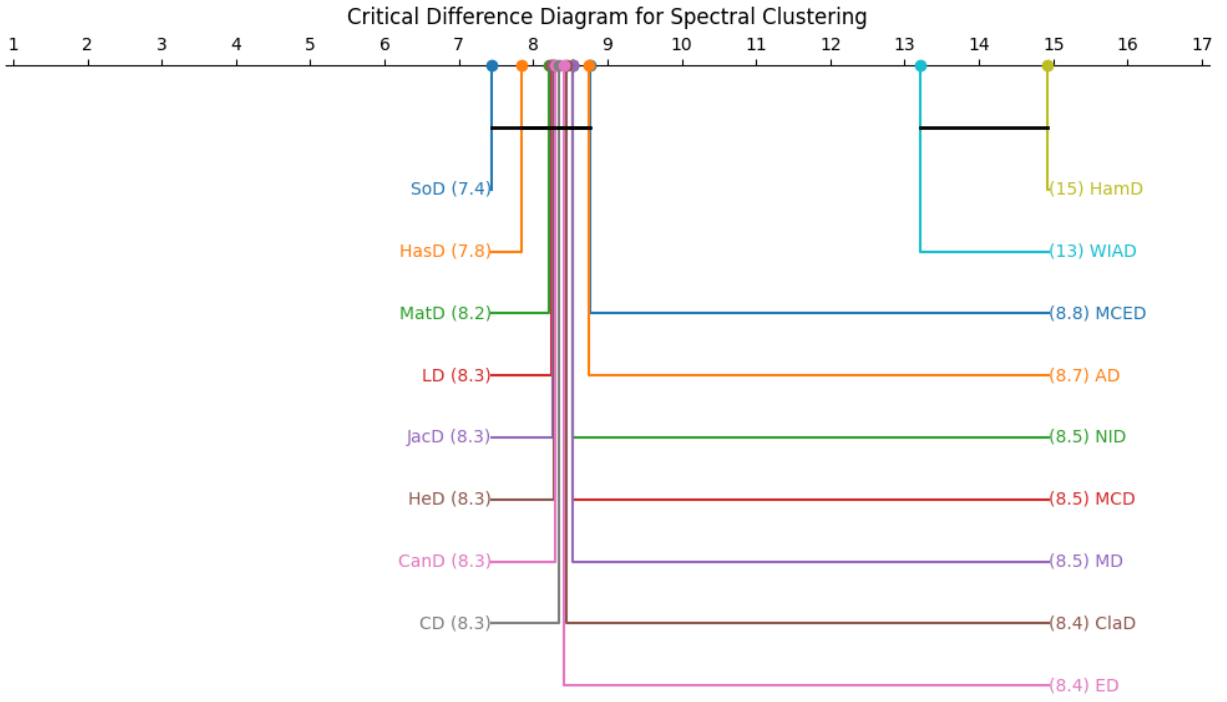
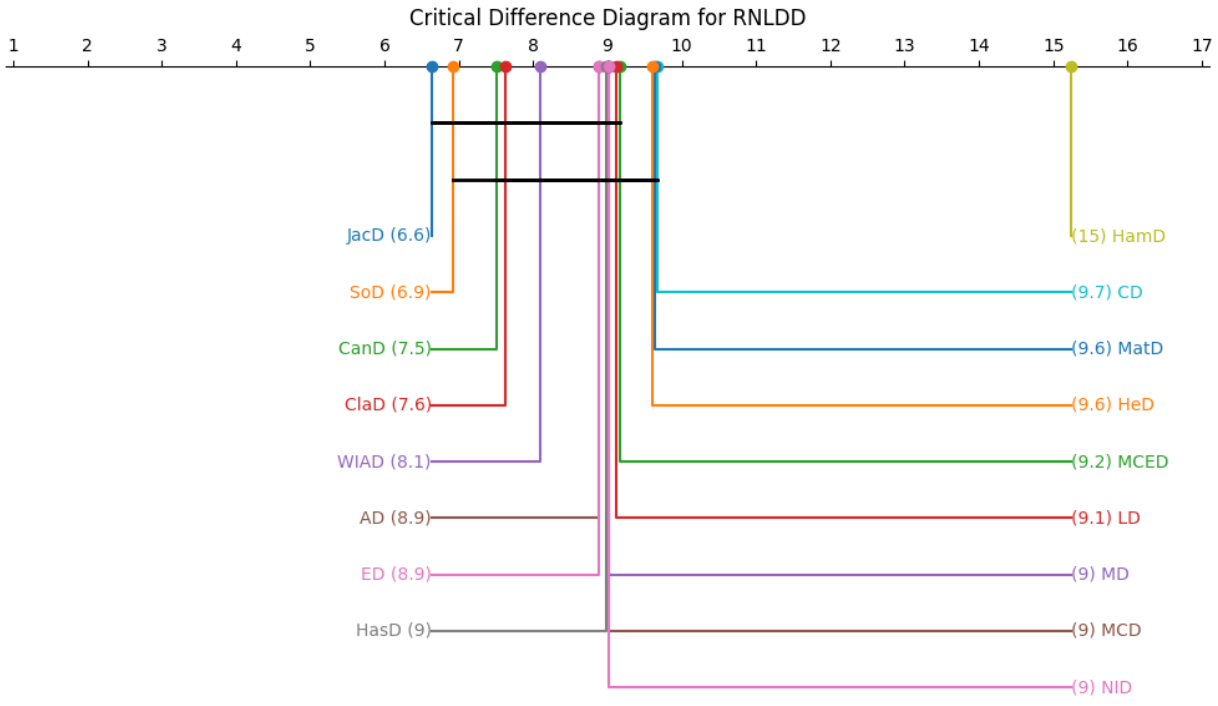
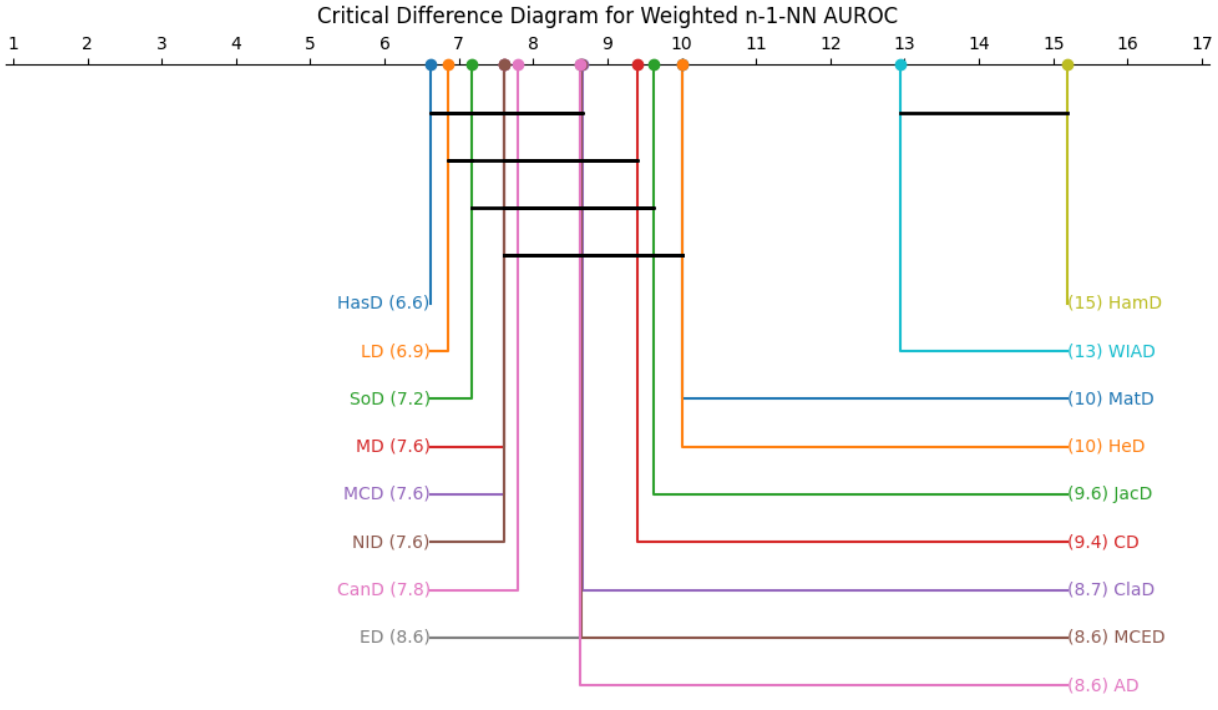


Figure 9: The critical difference diagram for "spectral clustering with ARI" diagnostic. Each measure's average rank over 81 datasets is given. The black bars are statistically indistinguishable groups.



(a) RNLDD with AUC Diagnostic. RNLDD is a local density-based algorithm.



(b) Leave-One-Out Weighted $(n-1)$-NN with AUC Diagnostic

Figure 10: The critical difference diagrams for two structural diagnostics using the AUC performance metric. Each subfigure shows the average measure rank over 81 datasets. The black bars are statistically indistinguishable groups.

5. Discussion

Recall that a diagnostic is compatible with a dataset if we can reject the joint-null hypothesis (HMP$\leq$ 0.05). In other words, after considering all measures, the diagnostic likely performs better than the relevant baseline (random chance or majority-label guessing). In Figure 1, we see that all diagnostics are compatible with at least 59/81 (>71%) datasets. Aggregate clustering (59) and RNLDD (62) are compatible with the fewest datasets, while SVC (77) and Weighted (n-1)-NN (78) are compatible with the most. As for the other null-hypothesis testing, the global Friedman test that checks if rankings are reasonably far away from the expected average ranking, all diagnostics passed.

As demonstrated in Figure 2, MCC consistently yields more statistical power than accuracy for all k-NN tested (for $k \in \{1, 3, 5, 7, 0.02n, 0.05n, \sqrt{n}\}$). Figure 3 reflects on statistical power for each selected algorithm in our pipeline; here, only 1-NN with MCC and Weighted (n-1)-NN have more statistical power (measure-rank separability) than 1-NN with accuracy.

In Figure 4, we see that the top five MCC values for 1-NN are identical to those based on 1-NN accuracy. However, the other six diagnostics are more distant: only K-Medoids has *any* of the target-bag (1-NN with accuracy's top five measures) in its top three, and the first algorithm to see the entire target-bag is Weighted (n-1)-NN, achieving only after most measures (13/17) are seen.

Now let us consider a null hypothesis: that diagnostics have no similarity to 1-NN with accuracy. This dissimilarity would be apparently random. If we expect a uniform-random distribution in Figure 4 with 17 measures, we expect about 1/17 increase in the available y-axis for every step on the x-axis. That is, we expect $1/17 \approx 5.9\%$ of the target-bag to fill for every increase of the top X measures of the respective diagnostic (seen by line color). For example, we would expect to see about 50% of the target-bag once half the measures (8.5) are seen. Figure 4 paints a similar picture–the other diagnostics cluster around (8.5, 50). Additionally, other key points suggesting a random distribution such as $(0.4 \times 17 = 6.8, 40)$ and $(0.6 \times 17 = 10.2, 60)$ are near the center mass of these non-1-NN diagnostics. Finally, there doesn't appear to be consistency in one diagnostic being left or right of the center mass–RNLDD begins as the leftmost (most similar in early Top-X trials) and ends as the rightmost (least similar in late Top-X trials). The other diagnostic with high similarity in the early Top-X trials (seeing 60% of the target-bag by X=6), aggregate clustering, similarly becomes more dissimilar when compared to other diagnostics (100% at X=15).

Reflecting on the time series measures over the CDDs (Figures 5–10), Jaccard distance (JacD) performs best for 1-NN under both accuracy (average rank $\bar{r} = 7.4$) and MCC ($\bar{r} = 7.2$), while Soergel distance (SoD) is best for the support vector classifier ($\bar{r} = 7.3$). For the structural diagnostics (evaluated by AUC), the winner for the global-scale diagnostic, weighted $(n-1)$-NN, is Hassanat distance (HasD; $\bar{r} = 6.6$); while the local density diagnostic, RNLDD, is JacD ($\bar{r} = 6.6$), with SoD closely following as second ($\bar{r} = 6.9$). For unsupervised diagnostics, Hellinger (HeD) and Matusita (MatD) distances are K-medoids' *and* aggregate clustering's best performers (all with $\bar{r} = 7.2$); while SoD leads spectral clustering ($\bar{r} = 7.4$). At the bottom end, Hamming distance (HamD) is consistently in last or second-to-last place.

6. Conclusion

In conclusion, the "best" measure is clearly diagnostic-dependent. Rankings induced by leave-one-out (LOO) 1-NN with accuracy do not reliably transfer to other diagnostic objectives in our benchmarking. We recommend careful consideration of your learning downstream task before accepting benchmark results. This includes diagnostic-dataset compatibility: by considering fused p-values for each dataset, we find that LOO 1-NN is compatible with fewer datasets than LOO support-vector classifier or LOO weighted $(n-1)$-NN diagnostics.

As for future benchmarking driven by 1-NN, we recommend tracking MCC estimates and reporting these induced rankings alongside accuracy. We find that MCC consistently yields more Friedman statistical power than accuracy. In other words, MCC causes a more decisive rank separation among measures–potentially reducing statistically indistinguishable ambiguity. More broadly, we suggest running objective-informed rankings across multiple diagnostic families to provide a safer basis for measure selection.

Our results signal several directions for further research. First, analyses can be done in multivariate or multiclass settings. However, computational expense is a hurdle as using just one diagnostic, LOO 1-NN, benchmarking can take months of server compute. Another next step is incorporating more computationally expensive measures, such as kernel dynamic time warping. Furthermore, one could investigate partitioning more diagnostics in different ways (besides supervised/unsupervised/structural) to theorize the generalizability of the respective partitions.

Overall, our central takeaway is that measure benchmarking should be objective informed with a proper performance metric. One cannot assume 1-NN generalizes—despite the intuitive idea of “the closest data should typically have the same label".

Acknowledgments

Code Availability Statement

Please see https://github.com/jarrenbr/Beyond-1NN-Accuracy/ for our available code.

Data Availability Statement

We only use publicly available data.

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Relationships

There are no additional relationships to disclose.

Patents and Intellectual Property

There are no patents to disclose.

Other Activities

There are no additional activities to disclose.

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